PORTFOLIO PROJECT — SQL ANALYSIS

Bank Loan *Risk Analysis*

A comprehensive SQL-based analysis of 38,000+ loan records to evaluate borrower risk, repayment behaviour, and financial performance metrics.

SQL Server | 38K+ Records | Aditi Vairale | 2024

SCROLL

Key Performance Metrics

Core financial KPIs calculated from the Bank Loan dataset, including total applications, funded amounts, and repayment statistics.

38,576
Total Loan Applications
MTD: Dec 2021 | PMTD: Nov 2021

$435.8M
Total Funded Amount
Across all loan statuses

$473.1M
Total Amount Received
Recovered from borrowers

12.05%
Average Interest Rate
MTD vs PMTD tracked monthly

13.33%
Average DTI Ratio
Debt-to-Income indicator

$37.3M
Bank Profit (Good Loans)
payment received – funded amount

$7.6M
Loss Amount (Bad Loans)
Charged Off — irrecoverable

86.2%
Good Loan Rate
Fully Paid + Current loans

GOOD LOANS

86.2%

Applications	33,243
Funded Amount	$370.2M
Amount Received	$435.8M
Status	Fully Paid + Current

BAD LOANS

13.8%

Applications	5,333
Funded Amount	$65.5M
Amount Received	$37.3M
Status	Charged Off

LOAN STATUS BREAKDOWN

Status-wise Performance

Aggregated metrics grouped by loan status — including count, funded amount, amount received, average interest rate, and DTI.

LOAN STATUS	APPLICATION COUNT	TOTAL FUNDED	TOTAL RECEIVED	AVG INTEREST RATE	AVG DTI
Fully Paid	32,145	$351.4M	$411.6M	11.64%	13.17%
Current	1,098	$18.9M	$24.2M	15.09%	14.72%
Charged Off	5,333	$65.5M	$37.3M	13.88%	14.00%

🏆 TOP 5 GOOD CUSTOMERS (LOAN > $25,000 — FULLY PAID)

PURPOSE	LOAN AMOUNT	STATUS
Debt Consolidation	$35,000	• Fully Paid
Home Improvement	$33,600	• Fully Paid
Small Business	$30,000	• Fully Paid
Major Purchase	$28,000	• Fully Paid
Debt Consolidation	$26,500	• Fully Paid

⚠ TOP 5 BAD CUSTOMERS (LOAN ≥ $35,000 — CHARGED OFF)

PURPOSE	LOAN AMOUNT	STATUS
Debt Consolidation	$40,000	• Charged Off
Small Business	$38,500	• Charged Off
Home Improvement	$37,000	• Charged Off
Debt Consolidation	$36,200	• Charged Off
Medical	$35,000	• Charged Off

—— SQL QUERIES

Query Documentation

All analysis was performed using SQL Server. Below are the key queries written for this project.

TOTAL APPLICATIONS

```sql
-- Total loan applications
SELECT COUNT(id)
  AS Bank_Loan_Applications
FROM [Bank_loan Data];

-- MTD (Month-to-Date)
SELECT COUNT(id)
  AS MTD_Applications
FROM [Bank_loan Data]
WHERE MONTH(issue_date) = 12
  AND YEAR(issue_date) = 2021;
```

FUNDED & RECEIVED AMOUNTS

```sql
-- Total funded amount
SELECT SUM(loan_amount)
  AS Total_Funded_Amount
FROM [Bank_loan Data];

-- Total amount received
SELECT SUM(total_payment)
  AS Total_Amount_Received
FROM [Bank_loan Data];
```

INTEREST RATE & DTI

```sql
-- Average interest rate
SELECT ROUND(AVG(int_rate),4) * 100
  AS Average_Interest_Rate
FROM [Bank_loan Data];

-- Average DTI ratio
SELECT ROUND(AVG(dti),4) * 100
  AS AVG_DTI
FROM [Bank_loan Data];
```

GOOD VS BAD LOAN %

```sql
-- Good loan percentage
SELECT
 (COUNT(CASE WHEN loan_status
   IN ('Fully Paid','Current')
   THEN id END) * 100)
  / COUNT(id) AS Good_Pct
FROM [Bank_loan Data];

-- Bad loan percentage
SELECT
 (COUNT(CASE WHEN loan_status
   = 'Charged Off'
   THEN id END) * 100)
  / COUNT(id) AS Bad_Pct
FROM [Bank_loan Data];
```

PROFIT & LOSS CALCULATION

```sql
-- Bank profit (good loans)
SELECT
 SUM(total_payment)
  - SUM(loan_amount)
  AS Profit_Amount
FROM [Bank_loan Data]
WHERE loan_status IN
  ('Fully Paid', 'Current');

-- Bank loss (charged off)
SELECT
 SUM(loan_amount)
  - SUM(total_payment)
  AS Loss_Amount
FROM [Bank_loan Data]
WHERE loan_status = 'Charged Off';
```

TOP 5 BAD CUSTOMERS (WINDOW FN)

```sql
SELECT * FROM (
  SELECT id, loan_status,
    purpose, loan_amount,
    ROW_NUMBER() OVER
    (ORDER BY loan_amount DESC)
    AS rank
  FROM [Bank_loan Data]
  WHERE loan_status = 'Charged Off'
    AND loan_amount >= 35000
) AS ranked_loans
WHERE rank <= 5;
```

LOAN STATUS SUMMARY (GROUP BY)

```sql
SELECT
  loan_status,
  COUNT(id)              AS Emp_ID_COUNT,
  SUM(total_payment)     AS TOTAL_AMT_RECEIVED,
  SUM(loan_amount)       AS TOTAL_FUNDED_AMT,
  AVG(int_rate)          AS AVG_RATE_OF_INTEREST,
  AVG(dti * 100)         AS DTI
FROM [Bank_loan Data]
GROUP BY loan_status;
```

INSIGHTS

Key Findings

Business insights extracted from the analysis to support lending strategy and risk management decisions.

01

Strong Loan Recovery Rate

86.2% of all loans are classified as "Good" (Fully Paid or Current), indicating a healthy overall repayment culture in the dataset.

02

$7.6M in Irrecoverable Losses

Charged-off loans resulted in $7.6M in net losses. Borrowers with higher DTI and interest rates showed elevated default probability.

03

$37.3M Net Profit from Good Loans

The bank earned $37.3M in net profit from fully paid and current loans, driven by interest income above the principal amount.

04

Higher DTI = Higher Default Risk

Charged-off borrowers averaged a 14% DTI vs 13.17% for fully paid loans — a key early-warning indicator for risk assessment models.

05

Debt Consolidation Dominates

Both top good and bad customer segments show "Debt Consolidation" as the primary loan purpose — indicating it's the highest-volume and highest-risk category.

06

MTD/PMTD Trend Monitoring

Month-over-month tracking of applications, funded amounts, interest rates, and DTI enables proactive identification of shifting risk patterns.

TECHNOLOGY

Tools & Tech Stack

Technologies used to build this end-to-end loan risk analysis project.

- SQL Server
- T-SQL (Window Functions)
- Aggregate Functions
- GROUP BY / HAVING
- CASE WHEN Logic
- Subqueries
- ROW_NUMBER()
- Date Functions
- Microsoft Excel
- Power BI

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